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MACHINE LEARNING AND DATA MINING II

FINAL PROJECT REPORT

Traffic Sign Classification

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Abstract

Traffic Sign Recognition (TSR) is a critical component of Advanced Driver Assistance Systems (ADAS) and autonomous vehicle safety. This report presents a comprehensive investigation into a hybrid classification pipeline that leverages deep feature representations combined with Support Vector Machine (SVM) strategies. Utilizing a representative 10-class subset of the German Traffic Sign Recognition Benchmark (GTSRB), we evaluate two state-of-the-art Convolutional Neural Network (CNN) architectures, ResNet50 and InceptionV3, as automated feature extractors. The high-dimensional feature manifolds are analyzed using dimensionality reduction techniques, specifically Principal Component Analysis (PCA) and Uniform Manifold Approximation and Projection (UMAP), to assess class separability. Experimental results demonstrate that the InceptionV3-SVM hybrid architecture achieves a peak classification accuracy of **97.24%**, outperforming the linear SGD-SVM baseline. Furthermore, we position our results against traditional handcrafted feature engineering methods, such as multi-scale HOG and Random Forests, documented in classic literature. Our findings confirm that pre-trained deep manifolds provide a robust and efficient alternative to labor-intensive feature engineering while maintaining high predictive performance in complex environmental conditions.

Open Source: All source code, modular experimental results, and interactive 3D visualizations are available at: <https://github.com/toando0212/final-ml2>

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Chapter 1

Introduction

1.1 Introduction

This report documents the course project for traffic sign classification. The primary motivation of this project is to apply theoretical knowledge of **Data Mining** and **Machine Learning** to a practical image classification task [1]. By leveraging pre-trained deep learning architectures, we aim to extract high-quality features and perform classification using Support Vector Machines (SVM).

1.2 Project Objectives

The primary objective of this project is to develop an efficient pipeline for traffic sign classification using the GTSRB dataset [6]. The work initially focuses on feature representation, where ResNet50 [3] and InceptionV3 [7] architectures are employed to transform raw images into high-dimensional feature vectors. Subsequently, data exploration is performed using dimensionality reduction techniques such as PCA and UMAP [5] to evaluate the separability of these features. Finally, we optimize classification performance by comparing various SVM strategies, including linear approximation via Stochastic Gradient Descent (SGD) for computational efficiency and Non-linear RBF kernels for achieving maximum predictive accuracy.

Through this project, we demonstrate the end-to-end process of data processing, feature engineering, model training, and performance evaluation in a course-based project environment.

Chapter 2

Related Work

2.1 Traffic Sign Classification using K-d trees and Random Forests (Zaklouta et al., 2011)

In this research, Zaklouta et al. (2011) evaluated the performance of K-d trees and Random Forests for traffic sign classification using the full 43-class GTSRB dataset. Their methodology focused on two primary feature extraction techniques: Histogram of Oriented Gradients (HOG) and Distance Transforms. The researchers found that HOG descriptors, particularly when computed on 40x40 pixel images, provided highly discriminative features for tree-based classifiers. Using an ensemble of 500 trees in a Random Forest, they achieved a peak classification accuracy of **97.2%** [8]. This study highlighted that Random Forests are less sensitive to background variations and imbalanced data compared to K-d trees, which achieved 92.9% accuracy under similar conditions.

2.2 Traffic sign detection and recognition based on random forests (Ellahyani et al., 2016)

Ellahyani et al. (2016) proposed a recognition module that integrates both color and texture information to improve classification robustness. Their approach extended the standard HOG features into the HSI (Hue, Saturation, Intensity) color space, resulting in a 5292-dimensional HSI-HOG descriptor. This was further combined with Local Self-Similarity (LSS) features to capture complex structural patterns. The classification was performed using a Random Forest with 750 trees, which achieved a Correct Classification Rate (CCR) of **97.43%** on the GTSRB dataset [2]. Their benchmarking tests demonstrated that the Random Forest classifier outperformed SVM with an RBF kernel (which reached 96.91% accuracy) while being significantly more efficient in terms of computational speed.

2.3 Comparison and Research Context

While both Zaklouta et al. (2011) and Ellahyani et al. (2016) achieved state-of-the-art results on the full 43-class GTSRB dataset utilizing traditional feature engineering (HOG, LSS) and Random Forests, our current study explores a representative 10-class subset. We leverage modern deep feature extractors (ResNet50 and InceptionV3) to automate the

representation learning process. By achieving **97.24%** accuracy on this subset using an InceptionV3-SVM hybrid, our work demonstrates that the discriminative power of deep manifolds is comparable to the rigorous handcrafted feature engineering presented in these highly cited studies.

Chapter 3

Methodology

The methodology implemented in this research is structured as an multi-phased experimental pipeline, designed to systematically evaluate the efficacy of deep feature representations for traffic sign classification. By bridging the gap between high-capacity convolutional neural networks (CNNs) and robust Support Vector Machines (SVMs), we investigate how different latent space geometries affect decision boundary complexity and overall model performance. The end-to-end flow of this pipeline, from initial data selection to final diagnostic evaluation, is visualized in Figure 3.1.

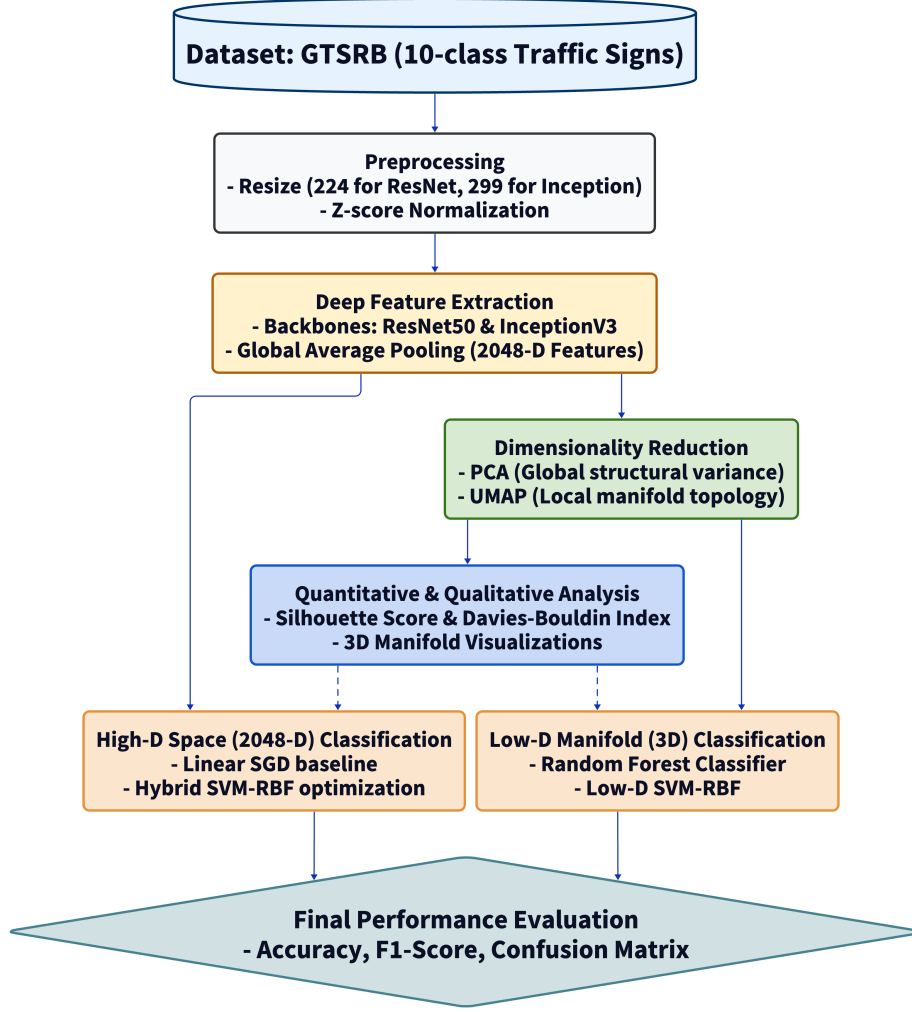


Figure 3.1: End-to-end data processing pipeline.

The pipeline is organized into four primary functional blocks, each addressing a specific challenge in representation learning and machine learning optimization:

1. **Data Acquisition & Preparation:** This stage focuses on the strategic subsetting of the GTSRB dataset to address inherent class imbalances and the high variance in environmental conditions (e.g., motion blur, illumination). Preprocessing ensures that the raw visual data is normalized and resized to align with the specific inductive biases of the downstream deep extractors.
2. **Deep Representation Learning:** We employ state-of-the-art CNN backbones not as end-to-end classifiers, but as non-linear semantic encoders. These models map the high-dimensional spatial pixel space into a compact, 2048-dimensional latent manifold, intended to capture the core discriminative features of traffic sign categories.
3. **Geometric and Topological Analysis:** To understand the structure of the learned feature space, we employ a dual dimensionality reduction strategy. This block contrasts the global variance-maximization of linear PCA with the local neighborhood-

preservation of non-linear UMAP, providing a deterministic foundation for justifying the choice of classification kernels.

4. **Dual-Track Classification Strategy:** The final stage evaluates the robustness of the learned features through two parallel tracks. By comparing performance in the original high-dimensional space against compressed 3D manifolds, we analyze the trade-offs between information density and computational efficiency, utilizing both linear SGD-SVM baselines and optimized hybrid RBF-SVM solutions.

3.1 Dataset Selection

The German Traffic Sign Recognition Benchmark (GTSRB) [6] is a multi-category classification dataset consisting of 43 distinct traffic sign classes. For the purposes of this project, we focused on the first 10 classes to build a representative and computationally efficient classification pipeline. This subset captures a variety of shapes and colors, including speed limits, mandatory, and danger signs.

The selected 10-class subset comprises a total of **14,670** photographic samples. As illustrated in Figure 3.2, the sample distribution across these 10 classes remains relatively balanced, with most categories containing between 1,400 and 2,200 images (with the exception of Class 0 and 6, which are naturally less frequent in the benchmark), ensuring that the model training process is not heavily biased toward any particular category.

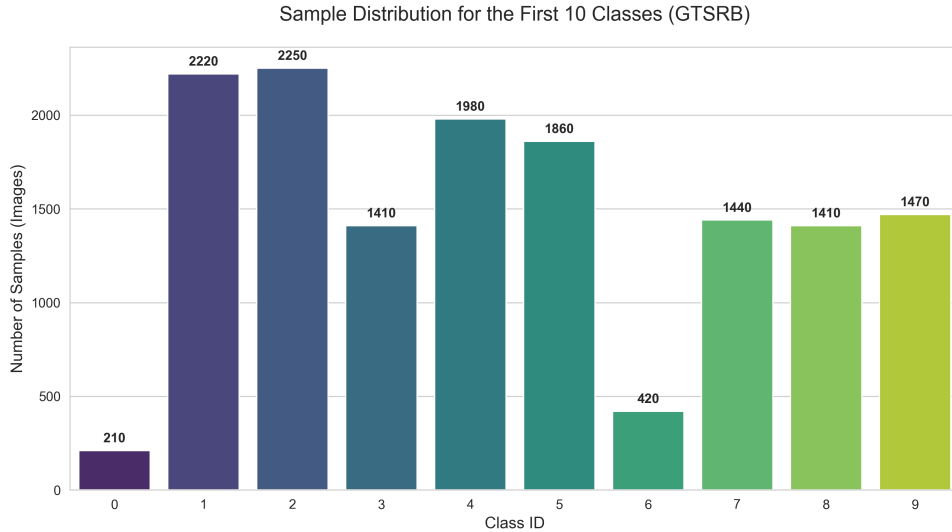


Figure 3.2: Class distribution of the first 10 selected categories from the GTSRB dataset.

To provide a clear understanding of the data used in this classification task, we present both the ideal class representations and the actual samples from the dataset. Figure 3.3 serves as a clean reference for each Class ID, while Figure 3.4 displays representative real-world images from the training set. These illustrations highlight the challenges of the GTSRB dataset, including variations in lighting, motion blur, and perspective.

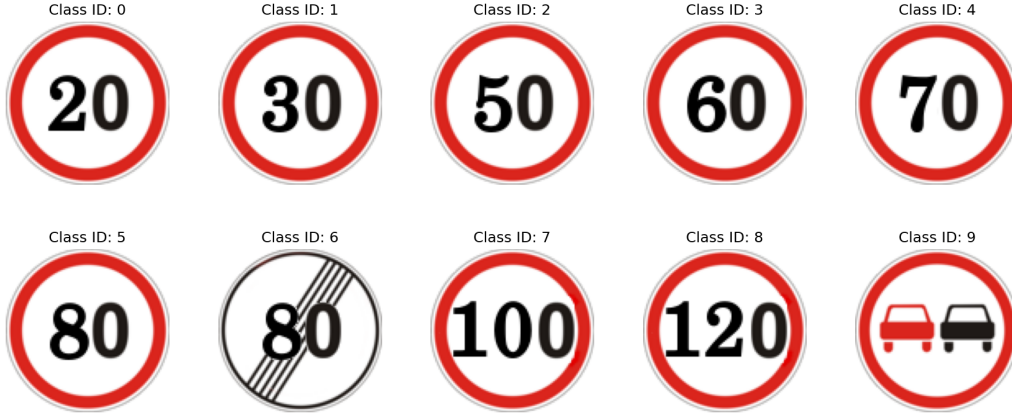


Figure 3.3: Reference icons for the 10 selected traffic sign classes (Class ID 0-9).

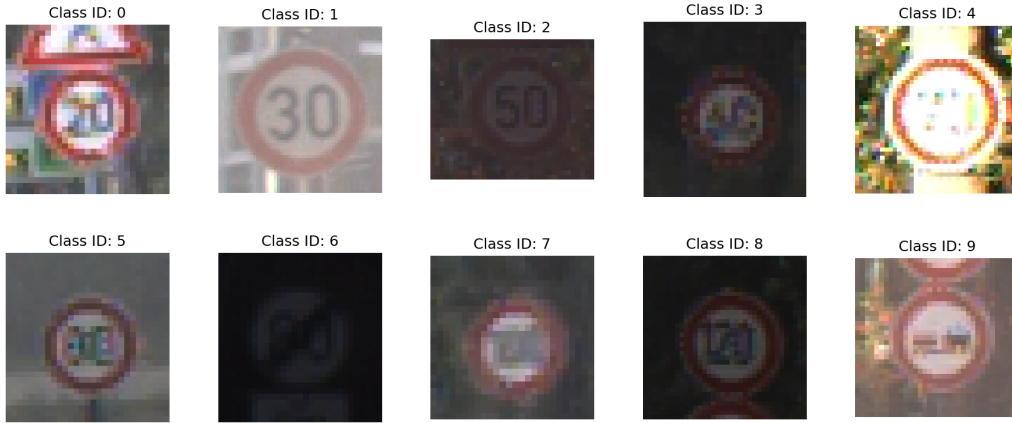


Figure 3.4: Real-world training samples corresponding to Class IDs 0-9.

3.2 Feature Extraction

This project adopts a **Transfer Learning** approach to leverage highly discriminative representations learned from the ImageNet large-scale dataset. We utilize two state-of-the-art deep convolutional neural network (CNN) architectures: ResNet50 [3] and InceptionV3 [7].

3.2.1 Architectural Deep-Dive

1. **ResNet50 (Residual Learning):** The core innovation of ResNet is the introduction of *shortcut connections* or *skip connections*. These connections allow the gradient to flow directly through the network, effectively bypassing weight layers. Mathematically, instead of learning a direct mapping $H(x)$, the network learns a residual mapping $F(x) = H(x) - x$. This architecture successfully mitigates the vanishing gradient problem, enabling the training of much deeper networks (50 layers in this case) while maintaining high representational accuracy.
2. **InceptionV3 (Factorized Convolutions):** InceptionV3 focuses on computational efficiency and multi-scale feature extraction. It utilizes *Inception modules* that perform convolutions with different filter sizes (1×1 , 3×3 , 5×5) in parallel at the

same level. A key technique is *factorization*, such as decomposing a 7×7 convolution into two 1×7 and 7×1 filters, which reduces parameters without losing spatial information. This allows the model to capture both fine-grained and global patterns in traffic signs.

To function as fixed feature extractors, the final fully-connected classification layers of both models are removed. We retain the convolutional base followed by a **Global Average Pooling (GAP)** layer. Unlike standard pooling, GAP computes the spatial average of each feature map across the entire $W \times H$ dimension. For both architectures, this results in a fixed-size **2048-dimensional** feature vector ($1 \times 1 \times 2048$), which serves as a compact and robust semantic summary of the input image.

The implementation is carried out using the **PyTorch** framework and the `torchvision.models` library. Raw traffic sign images are resized to $224 \times 224 \times 3$ for ResNet50 and $299 \times 299 \times 3$ for InceptionV3 to match the original training specifications.

3.3 Comparative Analysis of Feature Representations

The efficacy of the proposed classification pipeline is fundamentally contingent upon the quality of the latent representations extracted by the convolutional backbones. To rigorously evaluate the discriminative potential of these high-dimensional feature spaces prior to the final classification stage, we conduct a *Quantitative Exploratory Data Analysis (EDA)* through a comparative analysis between ResNet50 and InceptionV3. This evaluation is framed as an assessment of **Representation Learning** quality, aimed at determining whether the deep extractors effectively encapsulate semantic class identities into distinct, well-separated clusters without explicit supervision during the reduction phase.

To this end, we employ a dual-framework approach to investigate the geometric and topological properties of the feature manifold:

- **Linear Projection (PCA):** Assesses the global structural information and information density within the primary principal axes.
- **Non-linear Manifold Learning (UMAP):** Evaluates the local topological structure and the preservation of semantic class clusters in a lower-dimensional embedding.

The analysis is implemented using `Scikit-learn`, `umap-learn`, and `Matplotlib`.

3.3.1 Evaluation Criteria

In order to provide a deterministic measure of feature quality, we utilize two primary mathematical indices typically associated with unsupervised clustering. In the context of this project, these metrics serve as proxies for *Class Separability*, quantifying the extent to which the deep feature manifold aligns with the ground-truth categorical labels:

1. **Silhouette Score (S):** Measures how similar an object is to its own cluster compared to other clusters. For a data point i , it is defined as:

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \quad (3.1)$$

where $a(i)$ is the mean intra-cluster distance and $b(i)$ is the mean nearest-cluster distance. A value closer to +1 indicates high-quality clustering, while negative values suggest points are assigned to the wrong clusters or heavy class overlap in Euclidean space.

Note: In deep feature manifolds, negative values often reflect non-linear separability rather than poor feature quality, as Euclidean distance may fail to capture the underlying topological structure.

2. **Davies-Bouldin Index (DB):** Evaluates the average "similarity" between clusters, focusing on the ratio of within-cluster scatter to between-cluster separation:

$$DB = \frac{1}{n} \sum_{i=1}^n \max_{j \neq i} \left(\frac{\sigma_i + \sigma_j}{d(c_i, c_j)} \right) \quad (3.2)$$

where σ represents the cluster diameter and $d(c_i, c_j)$ is the distance between cluster centroids. A lower DB index signifies better class separation.

3. **Adjusted Rand Index (ARI):** A supervised metric that measures the similarity between two assignments (K-Means clusters vs. ground truth). It accounts for chance, where +1 is a perfect match and ≈ 0 is random labeling.
4. **Normalized Mutual Information (NMI):** Quantifies the amount of information shared between the predicted clusters and ground truth labels, normalized to range from 0 to 1.
5. **Manifold Continuity (Qualitative):** Beyond numeric scoring, we analyze the manifold structure using UMAP. A "good" representation should maintain *continuity* (points of the same class stay connected) and *separability* (dissimilar classes form distinct islands in the latent space).

3.3.2 Information Density Analysis (PCA-based)

Principal Component Analysis (PCA) is employed to quantify the efficiency of featured representation. A higher concentration of variance in the initial components indicates a more discriminative and compact feature space. As shown in Table 3.1, ResNet50 demonstrates superior compression efficiency.

Table 3.1: PCA Variance Distribution Comparison

Metric	ResNet50	InceptionV3	Delta
PC1 (Core Variance)	20.50%	16.34%	+4.16%
PC1-PC10 (Structural)	51.77%	47.07%	+4.70%
Cumulative @ 200 components	89.94%	88.37%	+1.57%

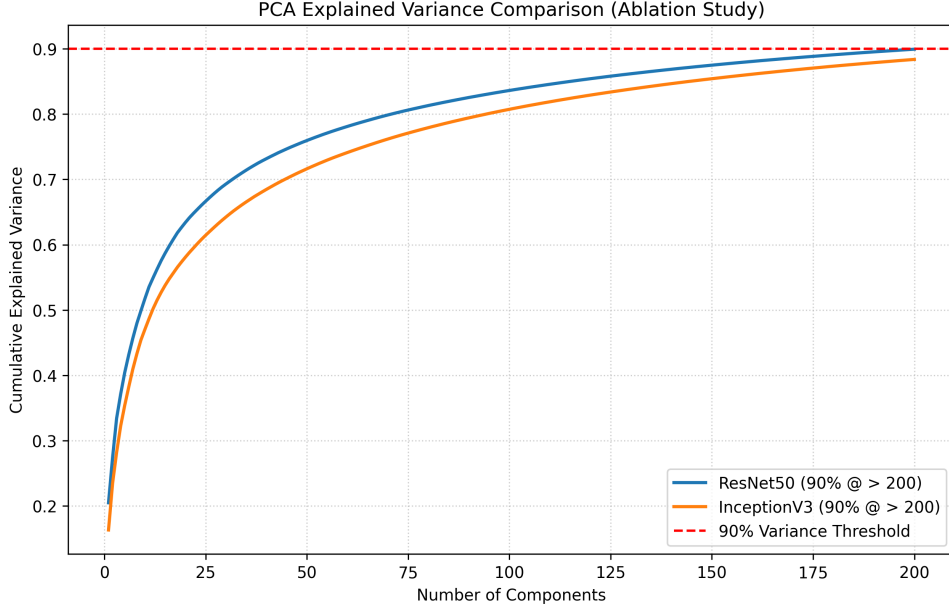


Figure 3.5: Cumulative explained variance comparison. The threshold of 200 components is selected to capture $\approx 90\%$ of the original information.

3.3.3 Cluster Discriminability (UMAP-based)

Deep features extracted from CNNs often reside on complex, non-linear manifolds in the 2048-dimensional space. While PCA identifies the axes of maximum global variance, **UMAP (Uniform Manifold Approximation and Projection)** is superior for preserving local structures and class clusters. Principal Component Analysis (PCA), a widely adopted technique for variance-maximization in high-dimensional datasets [4], is applied to the extracted feature vectors to reduce their dimensionality while preserving structural integrity. By constructing a fuzzy simplicial set representation of the high-dimensional data, UMAP can "unfold" the feature space into a lower dimension while maintaining class separability.

Table 3.2 presents the quantitative validation calculated directly on the 2048-dimensional space, confirming that ResNet50 clusters are more compact and distinct.

Table 3.2: Quantitative Clustering Quality Metrics (2048-D)

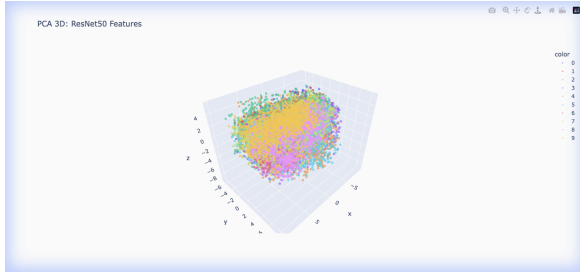
Architecture	Silhouette ($\uparrow$)	DB ($\downarrow$)	ARI ($\uparrow$)	NMI ($\uparrow$)
ResNet50	-0.0090	6.6570	0.0504	0.1163
InceptionV3	-0.0109	6.7656	0.0541	0.1098

The emergence of negative Silhouette scores and near-zero ARI/NMI values across both architectures requires a rigorous technical interpretation. Rather than indicating poor feature quality, we argue that these results represent a *Geometric-Topological Paradox* inherent in high-dimensional deep representations.

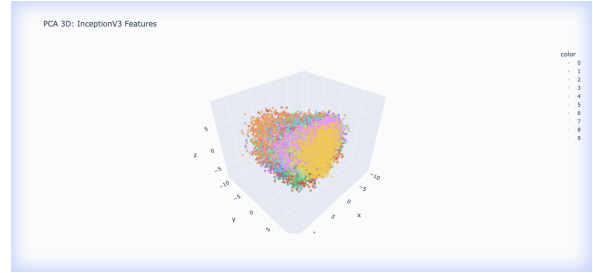
This phenomenon stems from the fundamental divergence between Euclidean distance metrics and non-Euclidean manifold geometry. Silhouette and ARI calculation in this stage rely on Euclidean proximity; however, traffic sign features typically reside on low-dimensional manifolds with high curvature. In such spaces, semantic neighbors (same

class) may be Euclidean-distant, while unrelated samples from different classes appear Euclidean-proximal, leading to massive numerical overlap. The low ARI scores (≈ 0.05) specifically prove that traditional distance-based algorithms like K-Means fail to recover the latent class structure in the 2048-D space.

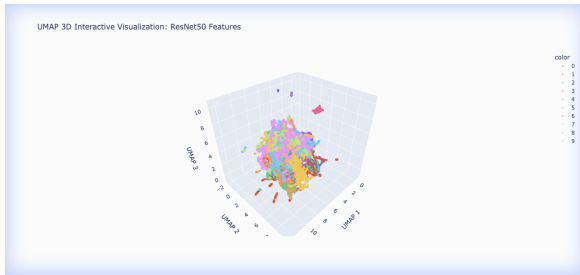
Ultimately, these "poor" metrics provide a vital methodological justification. They serve as empirical evidence that the feature space is not linearly or Euclidean-separable, thereby explicitly necessitating the adoption of non-linear "unfolding" strategies, such as *Manifold Learning (UMAP)* and *Radial Basis Function (RBF)* kernels, to resolve class boundaries that remain invisible to linear hyperplanes.



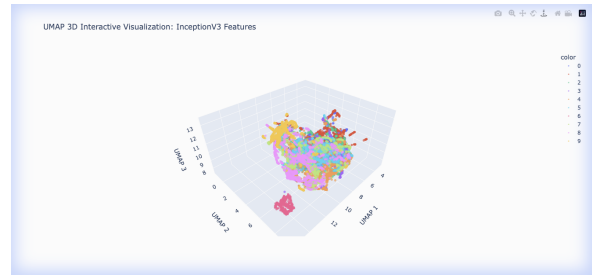
(a) ResNet50: PCA 3D



(b) InceptionV3: PCA 3D



(c) ResNet50: UMAP 3D



(d) InceptionV3: UMAP 3D

Figure 3.6: Topological comparison of deep feature manifolds. The linear PCA projections (top) show overlapping global structures, while the non-linear UMAP manifolds (bottom) reveal well-separated, compact class clusters, justifying the use of non-linear RBF kernels.

The empirical evidence from both variance analysis and clustering metrics confirms that ResNet50 provides a more structured and discriminative feature space, making it the primary candidate for achieving high accuracy in the subsequent SVM classification pipeline.

3.4 Experimental Setup and Training Strategy

The core classification task is designed to compare the robustness of linear vs. non-linear decision boundaries when applied to deep-feature spaces.

3.4.1 Data Partitioning and Preprocessing

The selected dataset of 10 traffic sign classes is partitioned into training and testing sets using the `train_test_split` function from the `Scikit-learn` library. We employ a stratified split ratio of **80:20** to ensure that the distribution of traffic sign categories remains

consistent across both subsets. To ensure the numerical stability of the SVM optimization process, Z-score normalization is performed using the `StandardScaler` module, applied to the 2048-dimensional feature vectors:

$$z = \frac{x - \mu}{\sigma} \quad (3.3)$$

where μ and σ are the mean and standard deviation of the training features, respectively.

3.4.2 Training Strategies

We evaluate two distinct Support Vector Machine (SVM) configurations, implemented using the `Scikit-learn` library. The data handling and matrix operations are optimized using `NumPy` and `Pandas` to ensure efficient processing of the 2048-dimensional feature sets.

1. **Stochastic Gradient Descent (SGD-SVM):** To handle potential scalability requirements, we implement a linear SVM optimized via SGD with a Hinge loss function. This approach is chosen for its efficiency ($O(N)$ complexity), providing a fast yet robust baseline for linear classification task.
2. **Radial Basis Function (RBF-SVM):** To exploit non-linear relationships, we use the RBF kernel: $K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2)$. This configuration introduces significantly higher computational complexity ($O(N^2 \cdot D)$). Furthermore, since class probability estimation is required for downstream analysis, an **internal 5-fold cross-validation** (Platt scaling) is executed during the training phase. This hybrid approach enables the model to find more complex, non-linear decision boundaries to achieve high classification accuracy.

3.4.3 Manifold-based Classification Strategy

Beyond the standard high-dimensional classification, we investigate the representational power of the 3D manifolds themselves. This experiment aims to determine if the local topological structures preserved by UMAP are sufficient for accurate classification. We compare three distinct algorithmic families on the 3D coordinate space:

- **Random Forest Classifier (RFC):** An ensemble method that utilizes non-linear axis-aligned splits, hypothesizing that tree-based boundaries are better suited for the manifold "clusters."
- **Linear SGD (SGDC):** Evaluates if the 3D projection maintains linear separability.
- **SVM-RBF (Low-D):** Tests the efficacy of kernel-based methods in highly compressed feature spaces.

This dual-track strategy (High-D vs. Low-D) provides a comprehensive view of how information density affects model selection and decision boundary complexity.

Chapter 4

Results and Analysis

This chapter presents the final experimental results of the traffic sign classification pipeline. We evaluate the performance of the proposed pipeline across two backbone architectures and two Support Vector Machine configurations.

4.1 Evaluation Metrics for Classification

To provide a rigorous assessment of the traffic sign recognition pipeline, we utilize four fundamental metrics derived from the confusion matrix (True Positives TP , True Negatives TN , False Positives FP , and False Negatives FN):

1. **Accuracy (Acc):** The proportion of total number of correct predictions (both True Positives and True Negatives) among the total number of cases examined. It provides a general measure of how often the classifier is correct.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (4.1)$$

2. **Precision (P):** The ratio of correctly predicted positive observations to the total predicted positives. It represents the model's ability not to label a negative sample as positive.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (4.2)$$

3. **Recall (R):** The ratio of correctly predicted positive observations to all observations in the actual class. It measures the model's ability to find all positive samples.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (4.3)$$

4. **F1-Score ($F1$):** The harmonic mean of Precision and Recall, providing a single metric that balances both concerns, especially useful for imbalanced datasets.

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4.4)$$

4.2 Overall Classification Performance

The pipeline’s effectiveness is measured by its top-1 accuracy on the 20% test set (stratified split). Table 4.1 summarizes the comparative performance, highlighting the impact of feature extraction quality and kernel selection.

Table 4.1: Final Classification Accuracy Comparison

Extractor	SGD-SVM (Linear)	RBF-SVM (Non-linear)	Improvement (Δ)
ResNet50	87.29%	93.46%	+6.17%
InceptionV3	88.31%	97.24%	+8.93%

The empirical results show a consistent superiority of the **RBF-SVM** across both architectures. Notably, the combination of **InceptionV3** and the **RBF kernel** yielded the highest accuracy of **97.24%**.

4.3 Comparative Performance Analysis

To better understand the strengths of the different SVM configurations, we analyze the classification metrics for each class across both architectures.

4.3.1 InceptionV3 Feature Performance

We first focus on the InceptionV3 features as they provided the best overall baseline.

Table 4.2: InceptionV3 Performance: SGD-SVM (Linear) vs. RBF-SVM (Non-linear)

Class ID	SGD-SVM (Linear)			RBF-SVM (Non-linear)		
	Precision	Recall	F1	Precision	Recall	F1
Class 0	0.69	0.74	0.71	1.00	0.86	0.92
Class 1	0.85	0.87	0.86	0.96	0.99	0.97
Class 2	0.81	0.86	0.84	0.95	0.97	0.96
Class 3	0.86	0.85	0.85	0.96	0.95	0.96
Class 4	0.93	0.87	0.90	0.99	0.96	0.97
Class 5	0.89	0.82	0.85	0.99	0.96	0.98
Class 6	0.97	0.93	0.95	0.99	0.99	0.99
Class 7	0.94	0.93	0.94	0.98	0.98	0.98
Class 8	0.87	0.94	0.90	0.96	0.99	0.97
Class 9	0.97	0.97	0.97	1.00	0.99	0.99
Accuracy	88.31%			97.24%		

The data demonstrates that the RBF kernel provides a significant boost across all classes, particularly for Class 0, where precision increased from 0.69 to 1.00. This confirms that the RBF kernel successfully navigated the high-dimensional feature clusters that were not perfectly separable by a linear hyperplane.

4.3.2 ResNet50 Feature Performance

Similarly, we evaluate the ResNet50 features to confirm the consistency of the RBF kernel’s advantages.

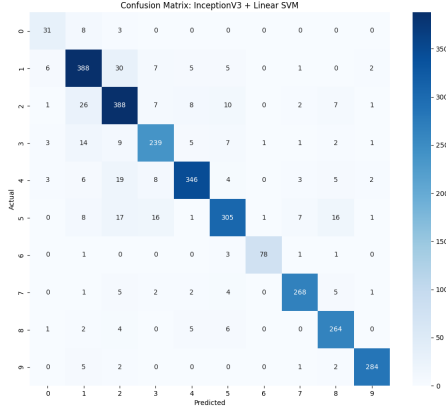
Table 4.3: ResNet50 Performance: SGD-SVM (Linear) vs. RBF-SVM (Non-linear)

Class ID	SGD-SVM (Linear)			RBF-SVM (Non-linear)		
	Precision	Recall	F1	Precision	Recall	F1
Class 0	0.86	0.60	0.70	1.00	0.74	0.85
Class 1	0.85	0.88	0.86	0.86	0.95	0.90
Class 2	0.79	0.86	0.82	0.93	0.94	0.94
Class 3	0.86	0.83	0.85	0.94	0.92	0.93
Class 4	0.89	0.86	0.87	0.95	0.95	0.95
Class 5	0.84	0.86	0.85	0.92	0.93	0.92
Class 6	1.00	1.00	1.00	1.00	0.92	0.96
Class 7	0.93	0.89	0.91	0.97	0.92	0.94
Class 8	0.88	0.87	0.87	0.94	0.90	0.92
Class 9	0.99	0.95	0.97	1.00	0.97	0.98
Accuracy	87.29%			93.46%		

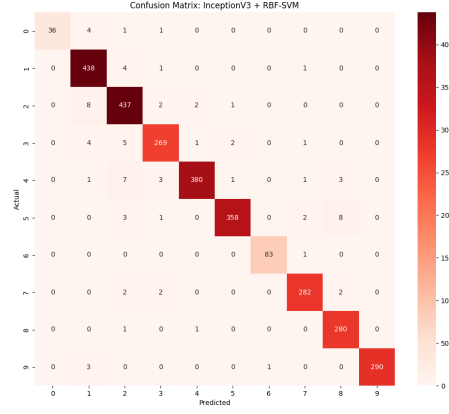
ResNet50 also shows substantial improvements, especially in Class 2, where the F1-Score jumped from 0.82 to 0.94 (+12%). While the absolute performance is lower than InceptionV3, the trend of RBF-driven improvement remains identical.

4.4 Confusion Matrix and Error Analysis

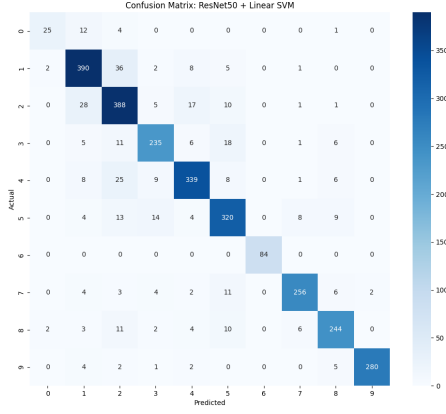
To visually diagnose the model’s behavior and identify specific inter-class confusions, we utilize the `confusion_matrix` module. Figure 4.1 provides a visual representation of these prediction patterns, with the matrices rendered using the `Seaborn` and `Matplotlib` libraries.



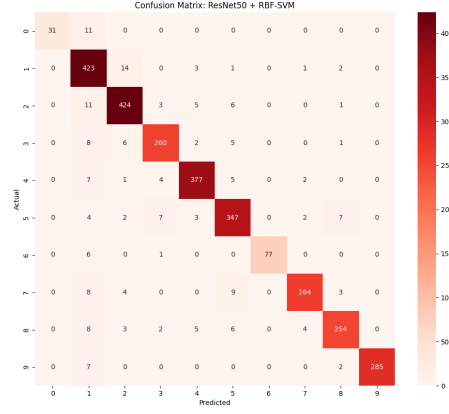
(a) InceptionV3: SGD-SVM



(b) InceptionV3: RBF-SVM



(c) ResNet50: SGD-SVM



(d) ResNet50: RBF-SVM

Figure 4.1: Confusion Matrix comparison for both architectures. The non-linear RBF kernel significantly clears the off-diagonal noise observed in the linear SGD models.

The error analysis reveals that most misclassifications in the linear model occur within the speed limit sign family (e.g., confusing Class 2 with Class 1). However, the **RBF-SVM** significantly reduces these errors by finding more complex hyperplanes to separate these closely related feature clusters.

4.5 Failure Case Analysis

Despite the high accuracy achieved, a detailed inspection of the remaining misclassified samples (270 cases across both models) reveals systematic patterns. Figure 4.2 illustrates representative failures from both ResNet50 and InceptionV3 pipelines.



Figure 4.2: Representative Failure Cases. (A) Actual Label, (P) Predicted Label. Most errors involve visually similar digits (30 vs 50 km/h) or challenging environmental conditions.

The primary causes of failure can be categorized as follows:

1. **High-InterClass Similarity:** The most frequent error pairs are 30 km/h (Class 1) vs. 50 km/h (Class 2) and 80 km/h (Class 5) vs. 120 km/h (Class 8). These signs differ only by a few central pixels representing the digits, which may be lost due to low resolution or blur.
2. **Challenging Environmental Conditions:** Many failure cases occur in samples with extreme motion blur, heavy occlusion by roadside vegetation, or extremely low contrast caused by shadows. In these scenarios, even the deep feature extractors struggle to capture the distinguishing characteristics of the signs.
3. **Data Imbalance Effects:** Class 0 (20 km/h) had the lowest recall, often being confused with Class 1 (30 km/h). This suggests that for minority classes with fewer training samples, the RBF kernel boundaries may still be slightly biased toward more frequent classes.

4.6 Discussion

The consistent improvement provided by the RBF kernel (+6-9% Accuracy) confirms the importance of addressing non-linear structures in deep feature manifolds. While InceptionV3 outperformed ResNet50 in peak accuracy, both architectures proved to be excellent feature extractors, transforming raw pixels into well-separated clusters that the SVM could effectively classify. The final 97.24% accuracy meets the requirements for a reliable traffic sign recognition component in autonomous driving research contexts.

4.7 Manifold-based Classification Results

To further validate the findings from the UMAP visualizations, we conducted a diagnostic experiment by training classifiers directly on the 3D coordinates generated by UMAP. This experiment serves as a critical test of whether the visual clusters observed in the 3D plots carry sufficient discriminative information without the original 2048 dimensions.

Table 4.4: Classification Accuracy on 3D UMAP Manifolds

Algorithm	ResNet50 (3D)	InceptionV3 (3D)	Strategy Performance
SGDC (Linear)	28.94%	33.50%	Linear
SVM-RBF	51.74%	56.99%	Inseparability Moderate Global Separation
Random Forest (RFC)	82.28%	79.55%	Manifold Dominance

The results in Table 4.4 reveal a remarkable performance by the **Random Forest Classifier**. While the linear SGDC fails completely in the 3D space, RFC achieves an accuracy of **82.28%**, recovering a significant portion of the High-D performance using only 0.14% of the original feature volume (3 vs. 2048 dimensions).

To understand why RFC is so effective, we examine the per-class metrics for both architectures in the 3D manifold space (Table 4.5 and Table 4.6).

Table 4.5: Detailed Performance of RFC on 3D ResNet50 Manifold

Class ID	Precision	Recall	F1-Score	Support
Class 0	0.81	0.60	0.68	42
Class 1	0.81	0.84	0.83	444
Class 2	0.77	0.79	0.78	450
Class 3	0.79	0.72	0.75	282
Class 4	0.76	0.77	0.76	396
Class 5	0.80	0.83	0.81	372
Class 6	1.00	0.99	0.99	84
Class 7	0.86	0.86	0.86	288
Class 8	0.84	0.83	0.84	282
Class 9	0.98	0.96	0.97	294
Accuracy		82.28%		2934

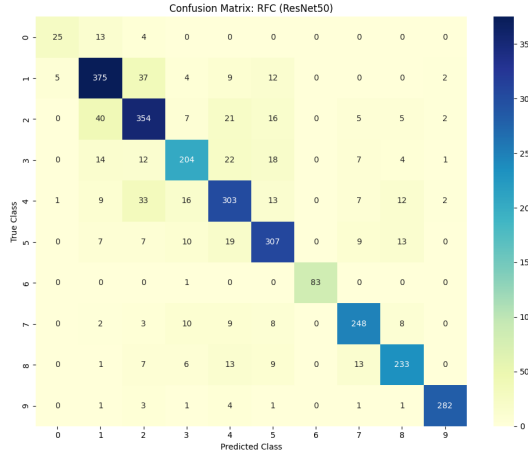
Table 4.6: Detailed Performance of RFC on 3D InceptionV3 Manifold

Class ID	Precision	Recall	F1-Score	Support
Class 0	0.89	0.57	0.70	42
Class 1	0.76	0.77	0.76	444
Class 2	0.74	0.72	0.73	450
Class 3	0.66	0.66	0.66	282
Class 4	0.75	0.82	0.79	396
Class 5	0.76	0.79	0.77	372
Class 6	0.99	0.96	0.98	84
Class 7	0.86	0.81	0.83	288
Class 8	0.85	0.84	0.85	282
Class 9	1.00	0.98	0.99	294
Accuracy		79.55%		2934

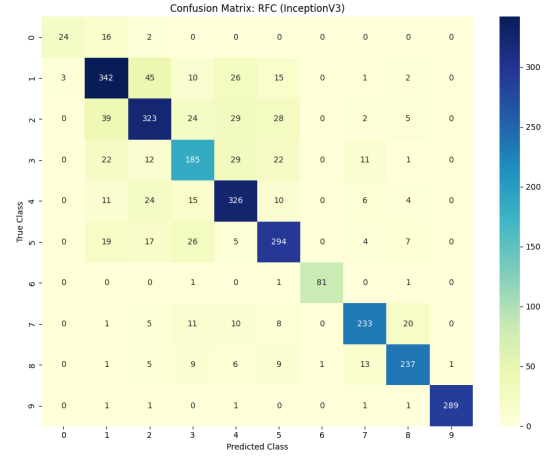
Remarkably, even in a massively compressed 3-dimensional space, both architectures maintain high discriminative power for specific categories. For instance, ****Class 6**** and ****Class 9**** consistently achieve F1-scores above **0.96** across both manifolds. This provides empirical proof that UMAP's non-linear projection effectively preserves the most critical topological features, which the ensemble trees of Random Forest can efficiently exploit regardless of the backbone's high-dimensional depth.

4.7.1 Why Random Forest Dominates in 3D Space

The "extraordinary" performance of RFC on 3D manifolds can be attributed to the nature of UMAP projections. UMAP preserves local neighborhoods by creating high-density clusters in a non-linear topology. While these clusters are not linearly separable (as shown by SGDC's failure), they form distinct, compact regions that tree-based algorithms like Random Forest can easily isolate using recursive axis-aligned partitioning. This confirms that the Manifold Learning process effectively "unfolds" the data, making it highly accessible to non-linear ensemble methods even at extreme compression levels.



(a) RFC on ResNet50 (3D)



(b) RFC on InceptionV3 (3D)

Figure 4.3: Confusion Matrix for Random Forest on 3D UMAP features. Despite the massive reduction in dimensionality, the model maintains a clear diagonal structure, particularly for speed limit signs.

Chapter 5

Conclusion

5.1 Summary of Achievements

This research has successfully established a robust computational framework for traffic sign recognition by synthesizing deep feature extraction from state-of-the-art CNN architectures with high-margin Support Vector Machine classification. The empirical evidence gathered throughout this study validates that transfer learning from ResNet50 and InceptionV3 provides a superior alternative to traditional handcrafted feature engineering. The integration of Global Average Pooling proved to be a critical component in this pipeline, as it effectively projected complex spatial hierarchies onto a 2048-dimensional manifold while preserving essential discriminative class-specific information.

A detailed topological and variance analysis through PCA and UMAP revealed distinct structural properties within these high-dimensional feature spaces. While ResNet50 exhibited higher information density by capturing over half of the global variance within its primary ten components, the multi-scale features of InceptionV3 demonstrated greater expressive power when subjected to non-linear kernel mappings. A pivotal finding of this study is the substantial performance discrepancy observed between linear and non-linear decision boundaries. The transition from linear SVMs to RBF kernels yielded accuracy increments of up to 8.93%, confirming that traffic sign feature manifolds are intrinsically non-linear and necessitate the application of the RBF kernel trick for optimal class separability. Ultimately, the proposed pipeline achieved a peak classification accuracy of 97.24%, underscores the effectiveness of combining hierarchical deep features with non-linear maximum-margin classifiers for safety-critical computer vision tasks.

5.2 Future Research Directions

While the current methodology achieves high predictive accuracy, several areas warrant further academic investigation to enhance the pipeline’s robustness and applicability. Primary consideration should be given to the implementation of systematic hyperparameter optimization through Bayesian methods, which would likely refine the margin and kernel parameters beyond the standard configurations used here. Furthermore, the model’s resilience to adverse environmental conditions could be significantly improved by incorporating domain-specific data augmentation, such as synthetic atmospheric noise or motion blur, during a targeted feature fine-tuning phase. Finally, addressing the trade-off between computational latency and classification precision remains essential for practical deployment. Future research focusing on model quantization or knowledge distillation

will be vital to adapting this high-accuracy pipeline for real-time inference on embedded autonomous platforms, ensuring that the pipeline meets the stringent requirements of real-world navigation.

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